Great! Press on.

MILESTONES

Sr. Project Design Phase
Karen Thurston
Fall 2000

MILESTONE	DATE DUE	DESCRIPTION
1	10/18	Request for Systems Services Form Problem Statement Matrix
2	10/18	Feasibility Assessment Report
3	10/23	Gantt Charts (2) Problems, Opportunities, Objectives & Constraints Matrix
4	10/25	E-R Diagram Entity/Definition Matrix
5	10/30	Data Flow Diagrams (Context, Event, Decomposition, System) Event-Response List 2 Decision Tables
6	10/30	High-Level Location Connectivity Diagram
7	11/3	Candidate Matrix Feasibility Matrix Systems Proposal Report
8	11/6	Physical Data Flow Diagram
9	11/13	Normalized Relations (3NF only w/ referential constraints) Complete Table Definitions (Table & Field Name; Field Name (PK), type, size, description, mask)
10	11/20	Input Prototypes (3) 1 screen, 1 doc + 1 Example Input filled-out (3) Output Prototype (2) 1 external, 1 internal report State Transition Diagram for Dialog Screens One Complete User dialog example
11	11/27	Partitioned DFD Structure Chart
12	12/1	Technical Report (handout)

MILESTONE EVALUATION SHEET

MILESTONE NO.

1

DATE: 10/18/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

TEAM NO. OR NAME	TEAM MEMBER
GREENSTONE CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

Greenstone

Greenstone
CONSULTING

Karen Thurston
Phone: (843)863-7308
E-mail:kthur106@aol.com

REQUEST FOR INFORMATION SYSTEM SERVICES

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
10/14/00	Business Department

SUBMITTED BY (key user contact)		EXECUTIVE SPONSOR (funding authority)	
Name	Dr. Bill Bowers	Name	
Title	Business Professor	Title	
Office	Jones Hall	Office	
Phone	(843) 863-7908	Phone	

TYPE OF SERVICE REQUESTED:

- ☐ Information Strategy Planning
- ☒ Business Process Analysis and Redesign
- ☒ New Application Development
- ☐ Other (please specify ______
- ☐ Existing Application Enhancement
- ☐ Existing Application Maintenance (problem fix)
- ☐ Not Sure

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

Charleston Southern University currently has a system of course evaluations that allows students to evaluate the course that they are taking as well as the professor who is instructing the course. The information gained through these evaluations is very valuable to the professor, the department, and to the overall academic success of the university. This information can be used to see which courses and professors are succeeding in their work and which could use improvement. The problem with the current system is that there are no ideal standards that can be used to compare each class to the ideal classroom situation as seen by the university and the members of each department. Dr. Bowers has requested a system be created that will allow students to evaluate each course and professor and then compare the results to the ideal scenario that has been previously established. With the new system, the data will be collected in such a way that it provides various statistical analysis reports that can give the university and the departments much more information than they have yet been able to receive from their current system of course evaluations.

BRIEF STATEMENT OF EXPECTED SOLUTION

The new system of course evaluations will use the Q Methodology (McKeown and Thomas, 1988) to collect and analyze students' responses. A set number of statements will be given and the student distributes the statements throughout several numbered columns whether or not the statement is most-like or least-like the class and professor that they are evaluating. The data will form a bell curve that can then be inputted into the computer system that will run the data through several statistical programs, creating useful reports of information for the university and the department to evaluate. From these reports it will be clear to see how each class or professor compares to the ideal classroom setting that has been established by Charleston Southern University.

ACTION (ISS Office Use Only)

- ☒ **Feasibility assessment approved** — **Assigned to** Karen Thurston
- ☐ **Feasibility assessment waived** — **Approved Budget** $ ______
- **Start Date** 10/14/00 **Deadline** 12/9/00
- ☐ **Request delayed** — **Backlogged until date:** ______
- ☐ **Request rejected** — **Reason:** ______

Authorized Signatures:

______ **Chair, ISS Executive Steering Body**

______ **Project Executive Sponsor**

PROBLEM STATEMENTS

PROJECT: Student Course Evaluation Analysis	**PROJECT MANAGER: Karen Thurston**
CREATED BY: Karen Thurston	**~~LAST UPDATED BY: Karen Thurston~~**
DATE CREATED: 10/14/00	**DATE LAST UPDATED: 10/14/00**

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1. Current system does not compare courses or professors to a set standard.	ASAP	High	Will set a standard for all classes.	1	Use Q Methodology that will have an ideal evaluation that is desired of the course and compare students' responses to the ideal.
2. ~~Data needs to be processed using statistical analysis to provide useful information to the university and the departments.~~	~~4 months~~	~~High~~	~~Will provide detailed analysis of the data.~~	2	~~Use the data, which forms a bell curve, and analyze it using several statistical equations.~~
3. Need an automated system to analyze course evaluations.	7 months	High	Will speed process and give accurate results with less effort.	3	Create automated system that will process the data and generate reports.

FIGURE 4.8

MILESTONE EVALUATION SHEET

MILESTONE NO.

2

DATE: 10/18/2000	COURSE NO.: COINS 497 ~~Senior Project Design~~
	INSTRUCTOR: Mr. Roberts

TEAM NO. OR NAME	TEAM MEMBER
GREENSTONE CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

Project Feasibility Assessment Report

I. Executive Summary

A. Summary of Recommendation

On September 15, 2000, Dr. Bill Bowers contacted Greenstone Consulting in regards to a project that would greatly enhance the current system of student course evaluations at Charleston Southern University. Greenstone Consulting will, using a new method of evaluations, create an automated system that will run statistical analysis of the data gathered from the evaluations. The new system will use Q Methodology to compare the students' views of the course and the professor to that of the university's ideal of what the course and professor should be. The new system will take this data, allow it to be inputted for each student, run it through several statistical equations, and produce reports that will contain valuable information to the university and each department.

B. Brief Statement of Anticipated Benefits

This new system, once implemented, will provide immediate benefits not only to the university, but also to each individual department, each professor, and, if made available, even to the students. The data will be analyzed to compare how different the course is to the ideal set by the university. Reports can be generated that also will show how similar the students views are in regards to a particular course. It will also show how different professors compare when teaching the same course, and how a professor's evaluations compare with each other. The major benefit will be that problems and successes can be easily spotted using exception reports and graphs, and trouble areas can be fixed for better overall academic success.

C. Brief Explanation of Report Contents

The contents of this report following this summary include a detailed description of the Student Course Evaluation project. It will include Greenstone Consulting, along with Dr. Bill Bowers, recommendations and ideas to solve the problems involved in the project. This report will also contain the project plan and schedule of each step of the project.

II. Background Information

A. Brief Description of Project Request

Dr. Bill Bowers of Charleston Southern University has requested of Greenstone Consulting that an automated computer system be created that can take the students' evaluations and produce useful reports for the university and the individual departments. The current system is not set up to compare each course and professor to a standard set by the university, and therefore does not give as complete information about where the weaknesses are and which areas need most improvement. The statistics involved with the new system would give the most accurate information possible that could be used for improvements of the courses as well as the professors. The final system should be complete and implemented by May of 2001. The goal of this system is to make the courses and operations at Charleston Southern University more effective by providing an accurate evaluation of each course and professor by the students of the university.

B. Brief explanation of the survey phase activities

Greenstone Consulting met with Dr. Bill Bowers of Charleston Southern University in September, 2000. Dr. Bowers will be the main contact for the project, and Karen Thurston will be the project manager for Greenstone Consulting. Karen will work with the assistance of Mr. Jim Roberts to design and implement this system as requested by Dr. Bill Bowers. The request was made by Dr. Bowers and evaluated by Karen Thurston. After evaluating the problems, opportunities, and objectives of this request, it has been decided by Greenstone Consulting to go forth with this request. An allotted time of eight months will be given to the completion of this project, with the final deadline being May, 2001.

III. Detailed Recommendation

A. Narrative Recommendation

1. Immediate fixes

Due to the fact that Student Course Evaluations are only performed once a semester, the university will continue with their current system of evaluating courses until the new system is completed and adopted by the university. If the system is accepted, it will be implemented, at the earliest, in the fall semester of 2001. Until that time there will be no immediate fixes to the system.

2. Enhancements

The enhancements made to the old system will be that of using Q-Methodology to evaluate the courses instead of only a few questions answered using 1-5 likertscale to rate most like or least like the course. The Q-Methodology will use 32 value statements that will be sorted amongst 9 categories, forcing the answers into a bell curve.

3. New Systems Development

Currently there is no automated way of reporting the results of the student evaluations. The new system will provide several options of analyzing the data, including sorting data by the mean for a course, calculating the correlations between students, calculating the correlation between different classes of the same professor, and various other ways. The new system will produce reports from the input that will reflect accurately the overall perception of the course and professor by the students that are enrolled in that course.

B. Project Plan

1. Initial Master Project Plan (Phase Level - Gantt Chart)

2. Detailed Plan for the Study or Definition Phase (Gantt Chart)

MILESTONE EVALUATION SHEET

MILESTONE NO.

3

DATE: 10/23/2000	**COURSE NO.: COINS 497** **~~Senior Project Design~~**
	INSTRUCTOR: Mr. Roberts

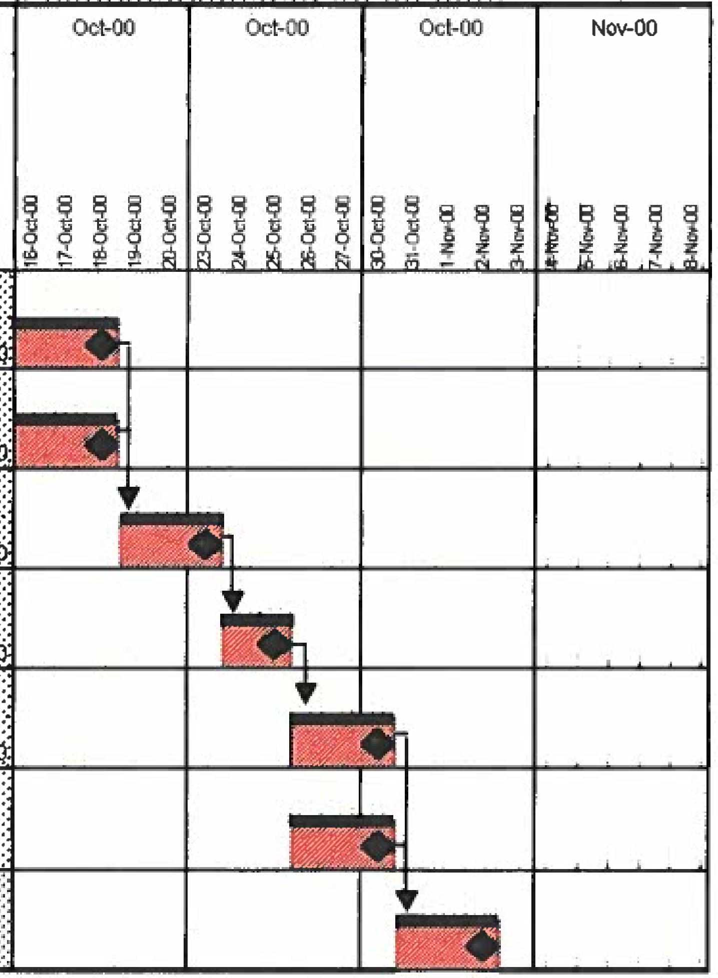
TEAM NO. OR NAME	TEAM MEMBER
GREENSTONE CONSULTANTS	Karen Thurston

Student comments:

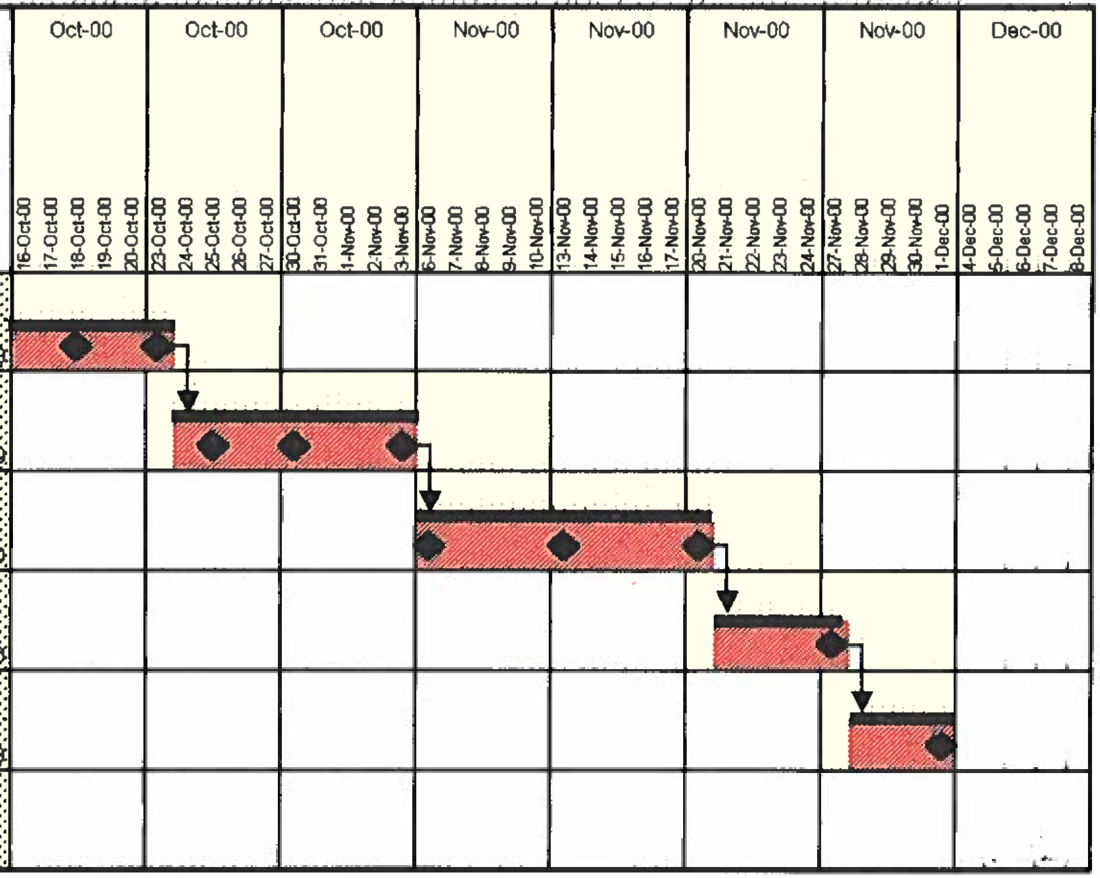
Instructor comments:

Score:

Step Number	Project Definition and Design Phase	Date Begin	Date End
1	Request for Systems Services Problem Statement Matrix	16-Oct-00	23-Oct-00
2	Feasibility Assessment Report	24-Oct-00	3-Nov-00
3	Gantt Charts Prob, Opp, Obj & Constr Matrix	6-Nov-00	21-Nov-00
4	E-R Diagram Entity/Definition Matrix	21-Nov-00	27-Nov-00
5	Data Flow Diagrams E-R List & Decision Tables	28-Nov-00	1-Dec-00
6	Locatin Connectivity Diagram		
7	Candidate & Feasibility Matrix Systems Proposal Report		



Step Number	Name	Date Begin	Date End
1	Project Definition	16-Oct-00	23-Oct-00
2	Design Phase	24-Oct-00	3-Nov-00
3	Build Phase	6-Nov-00	21-Nov-00
4	Test Phase	21-Nov-00	27-Nov-00
5	Document Phase	28-Nov-00	1-Dec-00



PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

Project:	**Student Course Evaluation Analysis**	**Project Manager:**	**Karen Thurston**
Created by:	**Karen Thurston**	**Last Updated by:**	**Karen Thurston**
Date Created:	**10/22/00**	**Date Last Updated:**	**11/9/00**

CAUSE AND EFFECT ANALYSIS		**SYSTEM IMPROVEMENT OBJECTIVES**	
Problem or Opportunity	**Causes and Effects**	**System Objective**	**System Constraint**
1. There is no automated program to generate reports from data collected by student course evaluations. 2. The current system of evaluations is not very thorough and does not give enough information to derive detailed reports.	1. The current system of evaluations uses a basic form with only a few questions. 2. The data received by these evaluations is not in a format that can be used for simple statistical calculations. 3. Reports need to be generated that give the university and the departments more useful data on each course and each professor, as seen by the students.	1. To use a new system of course evaluations that will implement Q-Methodology. 2. To retrieve the data from these evaluations and perform several statistical calculations on this data. 3. To create a system to automate the process of analyzing the results. 4. To produce reports that will reflect the students views on each course and professor compared to the benchmark established by the university.	1. The automated system must be able to be installed on any university PC with Windows 95 or higher. 2. The system should be user friendly and provide an input screen and several report choices based on the input. 3. This project must be completed by May 2001.

MILESTONE EVALUATION SHEET

MILESTONE NO.

4

DATE: 10/25/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

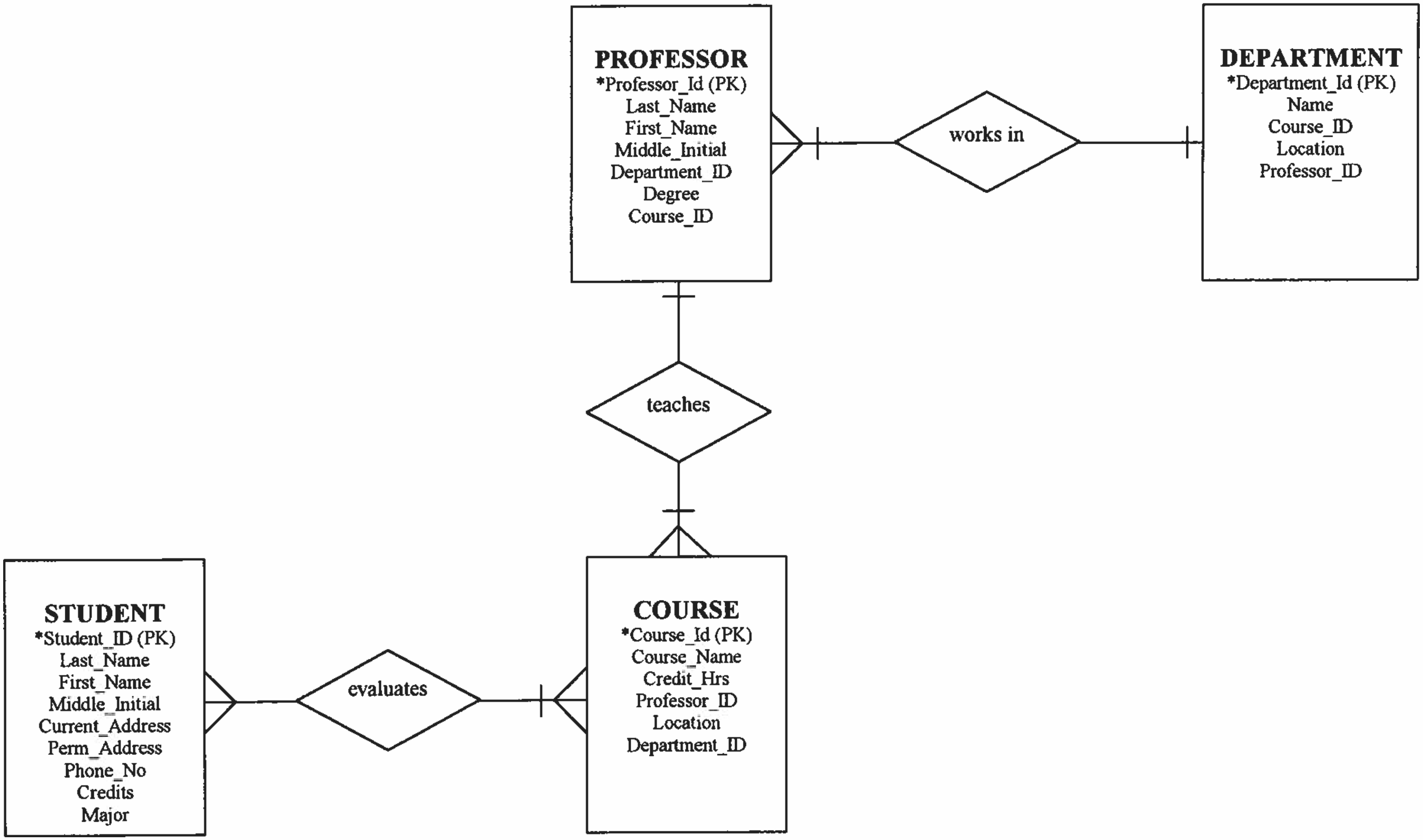
TEAM NO. OR NAME	TEAM MEMBER
Greenstone Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

Entity Relationship Diagram



Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION
STUDENT	A person enrolled in a course at the university.
COURSE	A course taught by the professor that has been approved by the university.
PROFESSOR	An employee of the university who works in a department and teaches a course.
DEPARTMENT	A department is comprised of professors who teach courses of a particular subject matter.

MILESTONE EVALUATION SHEET

MILESTONE NO.

5

DATE: 10/30/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

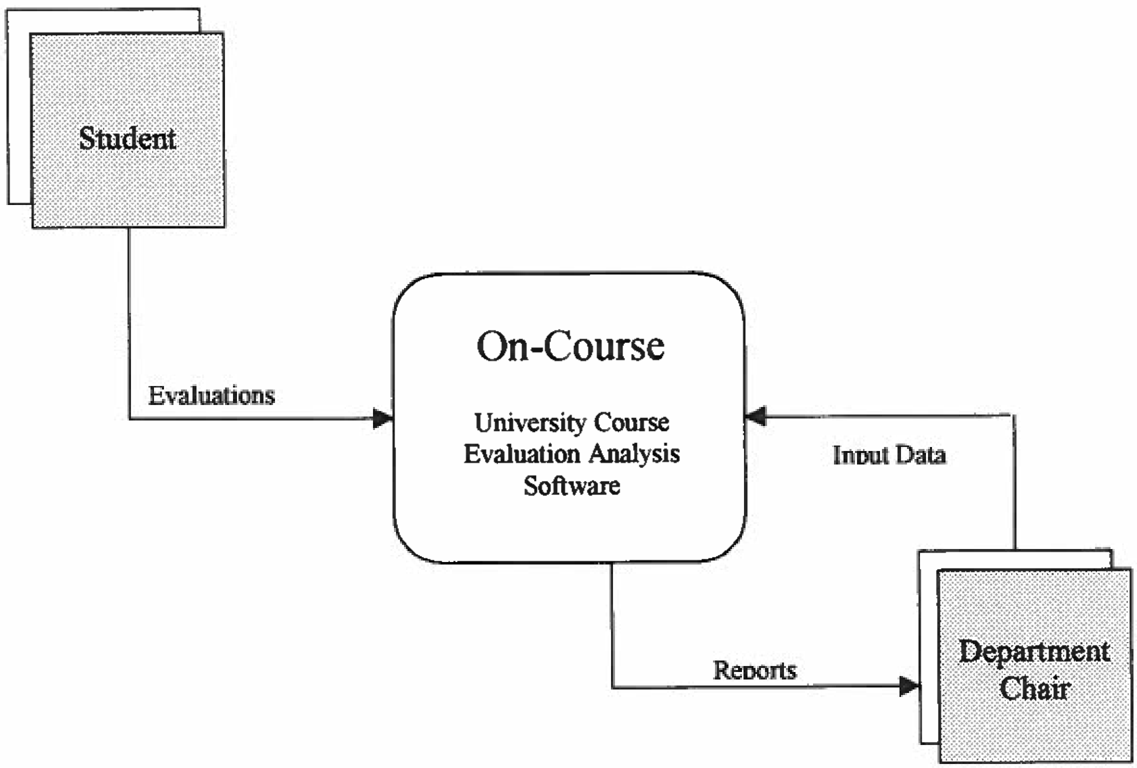
TEAM NO. OR NAME	TEAM MEMBER
Greenstone Greenstone CONSULTANTS	Karen Thurston

Student comments:

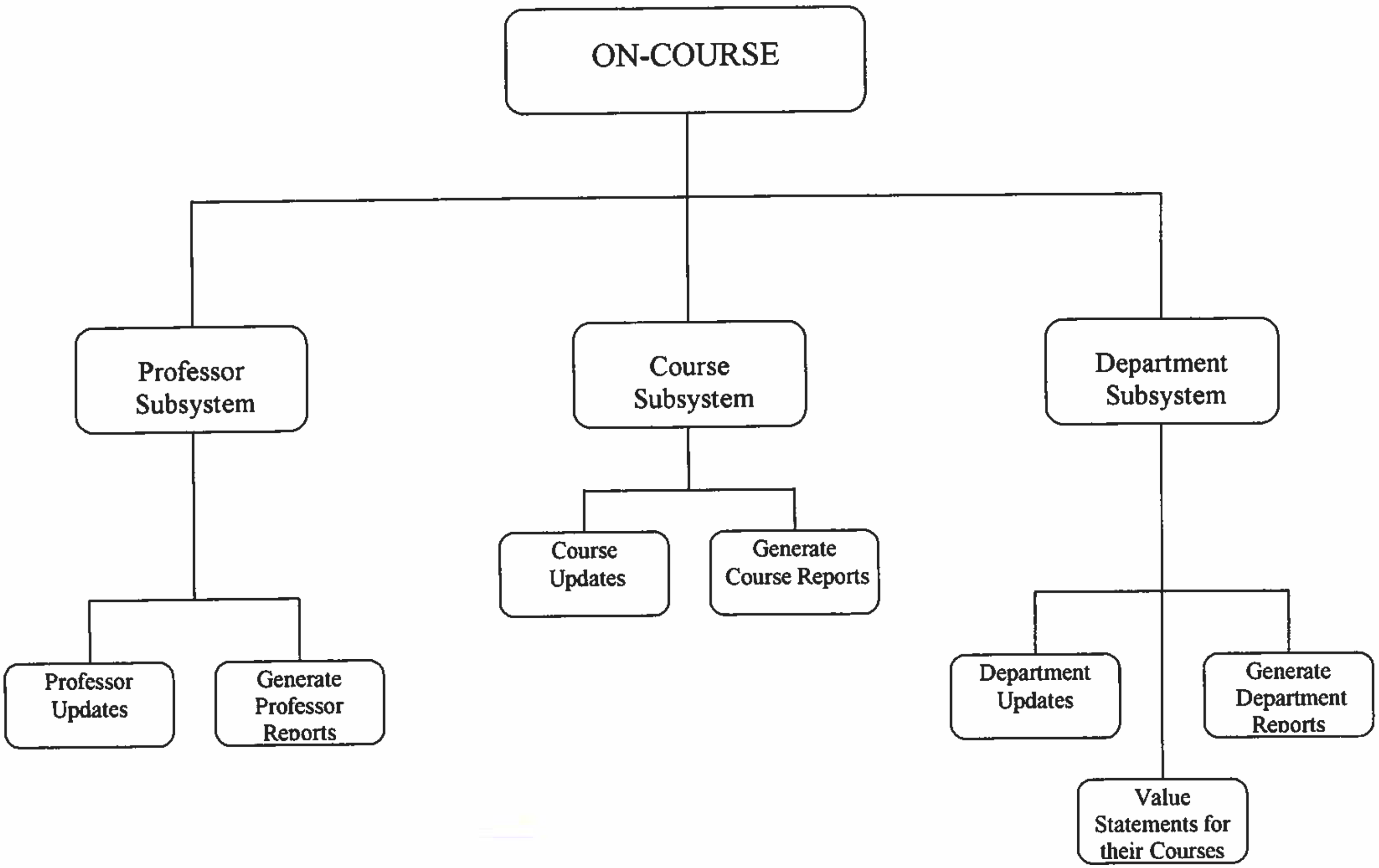
Instructor comments:

Score:

CONTEXT DIAGRAM

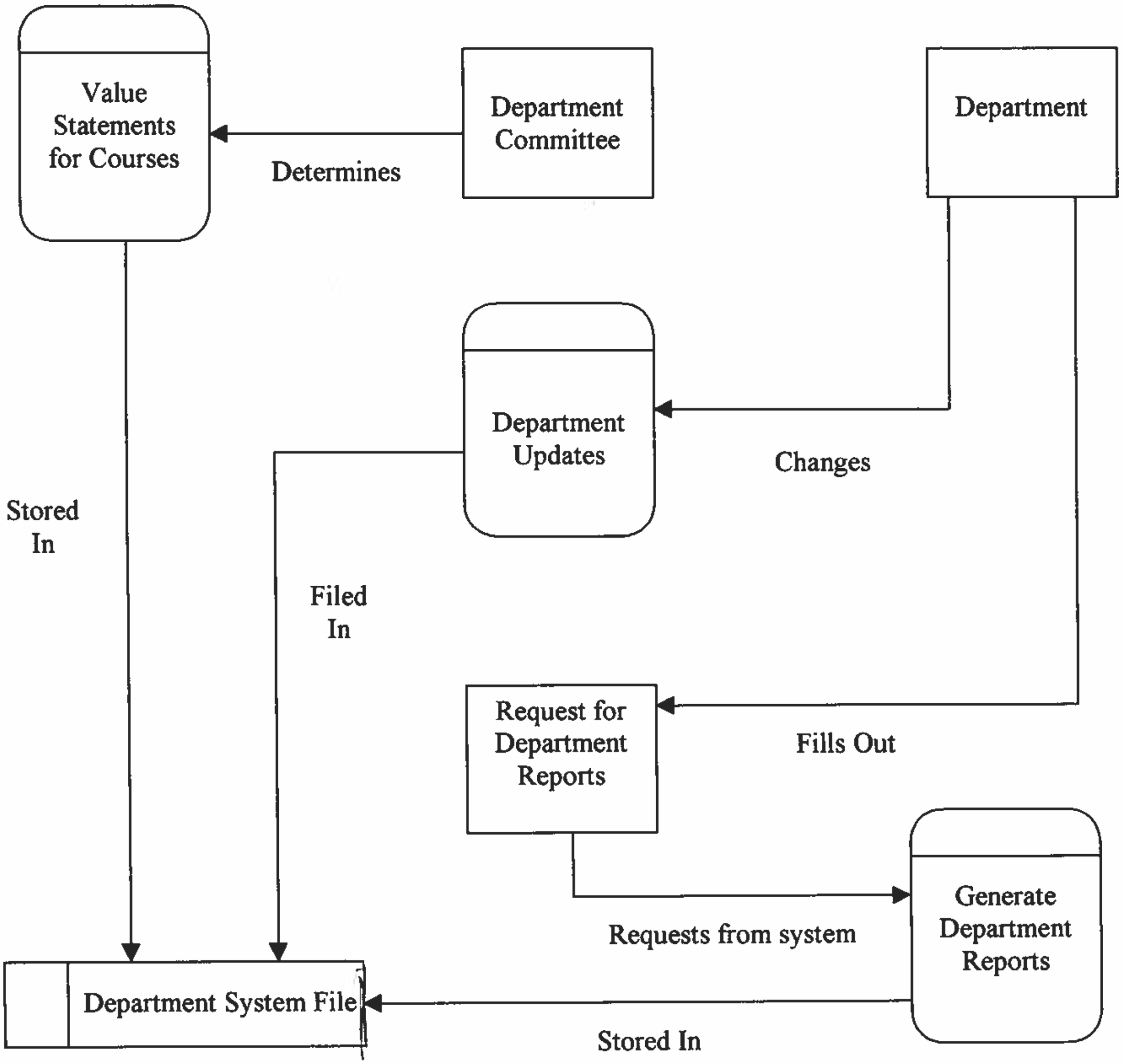


DECOMPOSITION DIAGRAM



? why is this here?

SYSTEM DIAGRAM



EVENT LIST

Event Description	Trigger (Inputs)	Responses (Outputs)
Student Evaluates Course	Data-Entry program stores their data	Updates that particular course database with the student's evaluation
Department Requests Analysis	Request made of the system to analyze the data for a course	Several reports are made available for printing or viewing
Course Report Generated	Analysis is requested for all evaluations for a particular course	Statistics for the evaluations of that course are given
Department Report Generated	Analysis is requested for all courses in a particular department	Statistics comparing all courses in a particular department are given
Professor Report Generated	Analysis is requested for all courses taught by a professor	Statistics comparing all courses taught by a professor are given
Courses Change	A course is added to the curriculum	The course database is updated to include the course
Departments Change	A department is added or information changes	The department database is updated
Professors Change	A professor is added or changed from a course	The professor database is updated accordingly
Value Statements are Changed	An authorized user (a department head) creates a new set of value statements	The value statements stored for that department are updated

Meaning the Number (currently 32) can change, The verbal descriptions can change, and/or the distribution across "piles" can change.

DECISION TABLE

Does the university need to be notified of a course evaluation?

Conditions	Rules			
	1	2	3	4
Did 10% or more of students give unsatisfactory evaluation?	Y	N	Y	N
Are there more than 5 averaged areas that need improvement?	Y	Y	N	N
Are there other courses taught by the professor that need improvement?	Y	Y	N	N
Actions				
University is notified	X			
University is not notified				X
Department is notified		X	X	

Not sure what this means...

Course	Prof.	Corr. Coeff
BUS 317-01	Bowers	−0.47
BUS 481-02	Long	−0.04
BUS 212-79	Smith	+0.03
BUS 332-01	Andersen	+0.93

Actual Avg vs. Ideal Avg.

Exception Report criteria?

Reports!

Available on-line (screen) as well as hard copy (paper).

DECISION TABLE

Evaluation Accepted and Analyzed for a Particular Course

Conditions	Rules		
	1	2	3
Is Evaluation completely filled out?	Y	N	N
Are all choices used once?	Y	N	N
Are all choices used only once?	Y	Y	N
Actions			
Evaluation is discarded		X	X
Evaluation is inputted	X		

Course	Inst.	# student Accept	# Reject	Total
		22	Ø	
		37	Ø	
		32	5	37

MILESTONE EVALUATION SHEET

MILESTONE NO.

6

DATE: 10/30/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

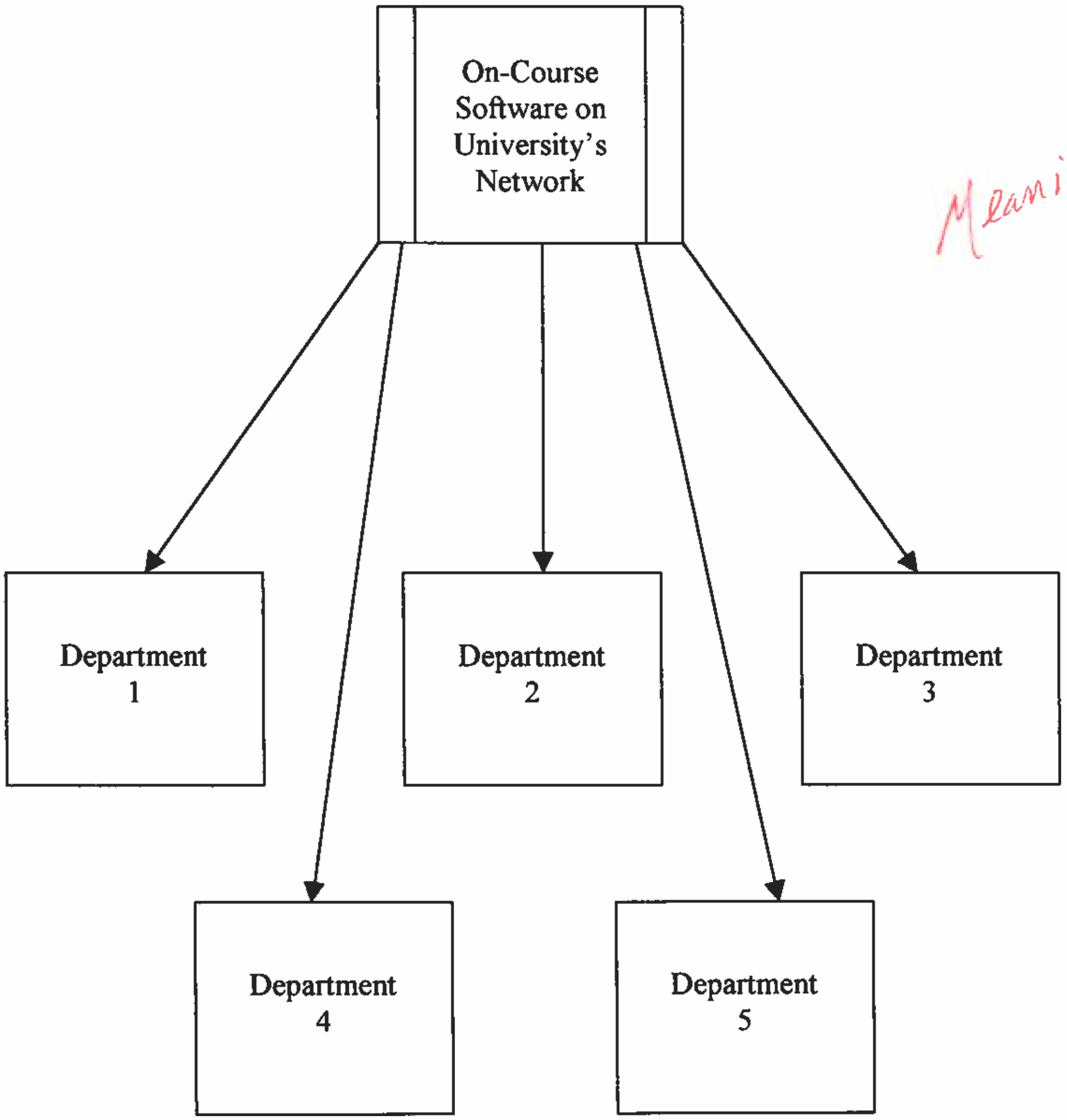
TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

LOCATION DIAGRAM



Meaning?

MILESTONE EVALUATION SHEET

MILESTONE NO.

7

DATE: 11/3/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

CANDIDATE MATRIX

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.	COTS software package designed to do basic statistical analysis of data.	After the data is manually inputted, the system would analyze the data and produce customized reports.	The system would have a computerized method of evaluating the courses and then would analyze the data and produce requested reports.
Benefits Brief description of the business benefits that would be realized for this candidate.	Would be a simplified and cost effective way to get the results.	Would be a customized program to give quick and informative reports that would be useful for the departments and the university.	Would eliminate the paperwork necessary for course evaluations and produce quick and informative reports that would be useful for the departments and the university.
Servers and Workstations A description of the servers and workstations needed to support this candidate.	Pentium Workstations that could load and run the software.	Pentium Workstations with Windows 95 or higher and a network that could run the software. ✓	Same as Candidate 2
Software Tools Needed Software tools needed to design and build the candidate (e.g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.	No Tools Needed, Software Package Purchased	Microsoft Access and Microsoft Excel with a Visual Basic Front End. If you say so.	Microsoft Access, Microsoft Excel, Microsoft Visual Basic and Java Programming.
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	Package Solution	Custom Solution	Custom Solution
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.	Batch	Same as Candidate 1	Same as Candidate 1
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).	Preprinted Forms developed by the software that report the results of the statistical equations. Printed on HP5SI LAN Laser Printers.	Customized forms that would print desired reports on HP5SI LAN Laser Printers.	Same as Candidate 2
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard, Mouse, Data Forms (Evaluations)	Same as Candidate 1	Keyboard, Mouse
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.	Evaluation data would be stored on the PC hard drive that it was input to.	All evaluations, course, and department information would be stored in the main database files. The data would be kept on the server, but also distributed through the department workstations on backup systems for safety and security precautions. ?	Same as Candidate 2

FEASIBILITY MATRIX

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility **Functionality.** A description of to what degree the candidate would benefit the organization and how well the system would work. **Political.** A description of how well received this solution would be from both user management, user, and organization perspective.	60%	Would be a pre-purchased software package and could not be customized for their needs. **Score: 60**	Would support the evaluations, department information, and course information. Would analyze the data and produce custom reports. **Score: 90**	Same as Candidate 2 with an computerized evaluation process added. **Score: 100**
Technical Feasibility **Technology.** An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. **Expertise.** An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	20%	Would be subject to upgrades from the pre-purchased software company. This Candidate would require a staff with the knowledge to install and maintain these upgrades. **Score: 60**	User friendly and designed for university faculty in mind. Would need minimal maintenance and could be upgraded upon request by Greenstone Consultants. **Score: 100**	Same as Candidate 2 only additional technical skill would be needed to maintain and upgrade the computerized evaluation system along with training and monitoring the usage. **Score: 80**
Economic Feasibility **Cost to develop:** **Payback period (discounted):** **Net present value:** **Detailed calculations:**	0%	Approximately $ Approximately $ Approximately $ **Score: unknown**	Approximately $ Approximately $ Approximately $ **Score: unknown**	Approximately $ Approximately $ Approximately $ **Score: unknown**
Schedule Feasibility An assessment of how long the solution will take to design and implement.	20%	Less than 3 months **Score: 100**	Less than 8 months **Score: 100**	More than 1 year **Score: 60**
Ranking:	**100%**	**68**	**94**	**88**

SYSTEM PROPOSAL

I. Introduction

A. Purpose of the report

The purpose of this report is to review and select a candidate solution from those that have been proposed for the system that has been requested. This report will give information and explanation for the proposed candidates and will recommend the best choice, as chosen by Greenstone Consultants.

B. Background of the project leading to this report

An analysis of the systems request has been done, and Greenstone has determined that there are several approaches to fulfilling the request. Three of the candidates have been identified in the candidate matrix. The information is based on preliminary work done by Greenstone to evaluate the system and the needs of this project.

C. Scope of the project

This project involves a system that will automatically evaluate the results of student course evaluations. The scope of the project could be expanded to include an automated system of evaluations or could be limited to simply an analysis for the current system of evaluations.

D. Structure of the report

This report will give the recommended solution to the system request based on the work done by Greenstone Consultants and the operational, technical, economic, and schedule feasibility of the candidate based on the needs of the university.

II. Tools and techniques used

A. Solution Generated

Having reviewed the three candidates and their feasibility, Greenstone is recommending candidate 2 as the best solution to the operational, technical, economic, and schedule needs of the university. The system needs to be customized to meet the needs of the university and to produce customized reports upon request from the departments or the university. The system also needs to be easily maintained and used by the faculty of the university. Although some customized systems would meet these requirements, it is also important that the system be economically feasibly and done in a reasonable amount of time.

B. Feasibility analysis (cost-benefit)

Candidate 2 will provide all the benefits of a system that is customized for the university's needs. It is the ideal candidate for a system that will analyze the student's course evaluations and produce reports that will benefit the professors, the departments, and the university. Candidate 2 will be created and supported by Greenstone Consulting. This system can be maintained and operated with very little technical knowledge necessary to the users. This candidate also complies with the economic feasibility of the university and can be completed and implemented within a reasonable amount of time.

III. Information systems requirements

This candidate would produce customized software that could be installed on any Pentium workstation or on a server and could be accessed from any of these workstations. The workstations could store the software as well as the databases needed, and would require connection to a LAN printer to produce the customized reports.

IV. Alternative solutions and feasibility analysis

The alternative solutions are found in candidates 1 and 3. These candidates propose systems that are either below the operational desires of the university or above the economic and schedule feasibility. Candidate 1 proposes a pre-packaged statistical system that could be purchased and used to run basic statistical calculations. This system would not produce the customized results that are desired, and would not be flexible to changes and upgrades from Greenstone. This system would be cost effective and timely, but the benefits would be so few that the cost would not outweigh it.

Candidate 3 would be an extensive system that would meet all the needs requested system. It would require advanced programming which would make it an expensive and time consuming project that somewhat outweighs the benefits that would come from this advanced system. It would also require more technical training and maintenance because there would be a wide scope of users for the computerized evaluation system.

V. Recommendations

Based on the analysis of the three candidates, Greenstone Consulting is recommending candidate 2 as the best system proposal to meet the needs of the university. Candidate 2 is the best combination of operational, technical, economic and schedule feasibility amongst the three candidates. It is a customized system that fine can be easily used and maintained and will provide the results that are desired.

Candidate 2 is the top choice by Greenstone to meet the needs of the university.

VI. Appendices

MILESTONE EVALUATION SHEET

MILESTONE NO.

8

DATE: 11/6/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

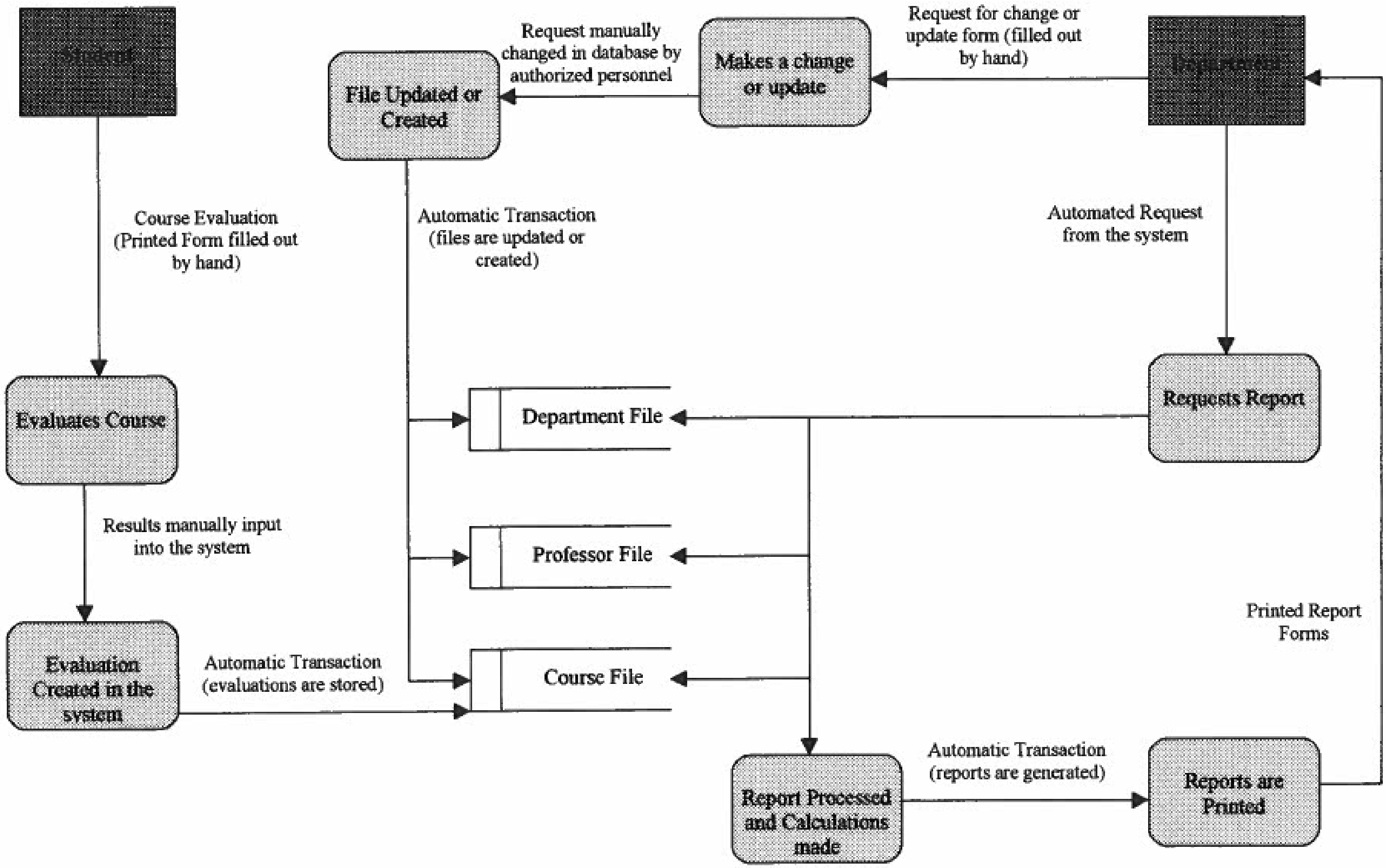
TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

PHYSICAL DATA FLOW DIAGRAM



MILESTONE EVALUATION SHEET

MILESTONE NO.

9

DATE: 11/13/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

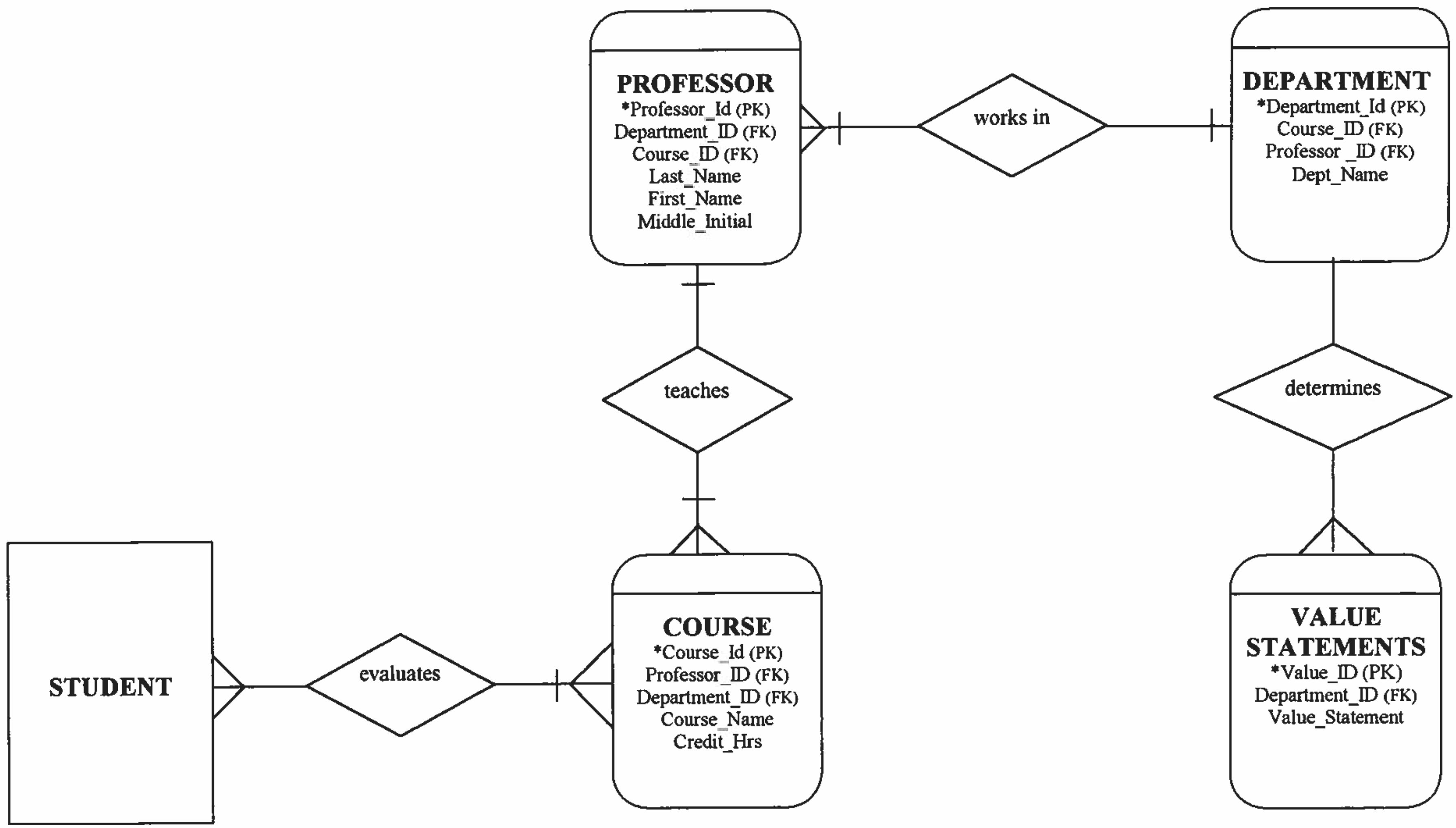
TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

Normalized Relations Diagram



COMPLETE TABLE DEFINITIONS

Table Name	Field Name	Type	Size	Description	Mask
Course	Course_Id (PK)	VARCHAR	12		AAAA-###-###
	Department_Id (FK)	VARCHAR	4		AAAA
	Professor_Id (FK)	CHAR	3		###
	Course_Name	VARCHAR	30		
	Credit_Hrs	CHAR	1		#
Professor	Professor_Id (PK)	CHAR	3		###
	Department_Id (FK)	VARCHAR	4		AAAA
	Course_Id (FK)	VARCHAR	12		AAAA-###-###
	Last_Name	VARCHAR	20		
	First_Name	VARCHAR	20		
	Middle_Initial	CHAR	1		A
Department	Department_Id (PK)	VARCHAR	4		AAAA
	Professor_Id (FK)	CHAR	3		###
	Course_Id (FK)	VARCHAR	12		AAAA-###-###
	Dept_Name	VARCHAR	20		
Value Statements	Value_Id (PK)	CHAR	2		##
	Department_Id (FK)	VARCHAR	4		AAAA
	Value_Statement	VARCHAR	50		

larger?

MILESTONE EVALUATION SHEET

MILESTONE NO.

10

DATE: 11/20/2000	COURSE NO.: COINS 497 Senior Project Design INSTRUCTOR: Mr. Roberts

TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANTS	Karen Thurston

Student comments:

Instructor comments:

Score:

Charleston Southern University

COURSE EVALUATIONS
Fall 2000

DEPARTMENT: BUSI

SECTION:

COURSE:

PROFESSOR: 123

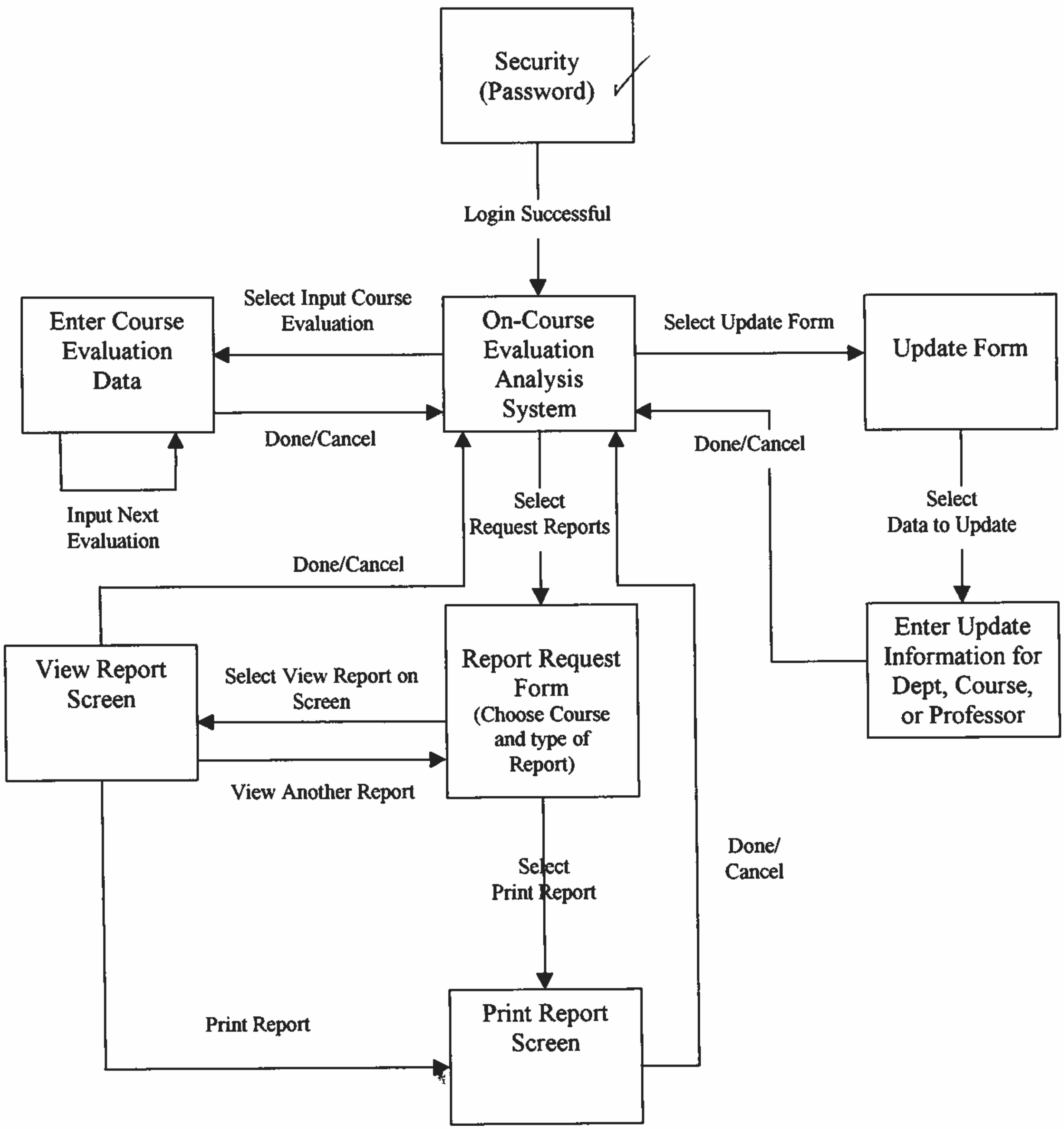
CORRELATION COEFFICIENT
of the
MEANS FOR EACH COURSE
as taught by
DR. BILL BOWERS

	BUSI 218	BUSI 317	BUSI 456	BUSI 610
BUSI 218	1			
BUSI 317	0.806566	1		
BUSI 456	0.827869	0.888231	1	
BUSI 610	0.877366	0.798002	0.825128	1

OUTPUT
PROTOTYPE

- INTERNAL

STATE TRANSITION DIAGRAM



Charleston Southern University

ON-COURSE

Evaluation Analysis System

Username:

Password:

EXAMPLE
INPUT
PROTOTYPES

Charleston Southern University

ON-COURSE

Evaluation Analysis System

EVALUATION INPUT FORM

Instructor # Choose One

Semester: Fall

Year: 1999

Course # Choose One

Department # Choose One

Section # Choose One

OK

Next Back

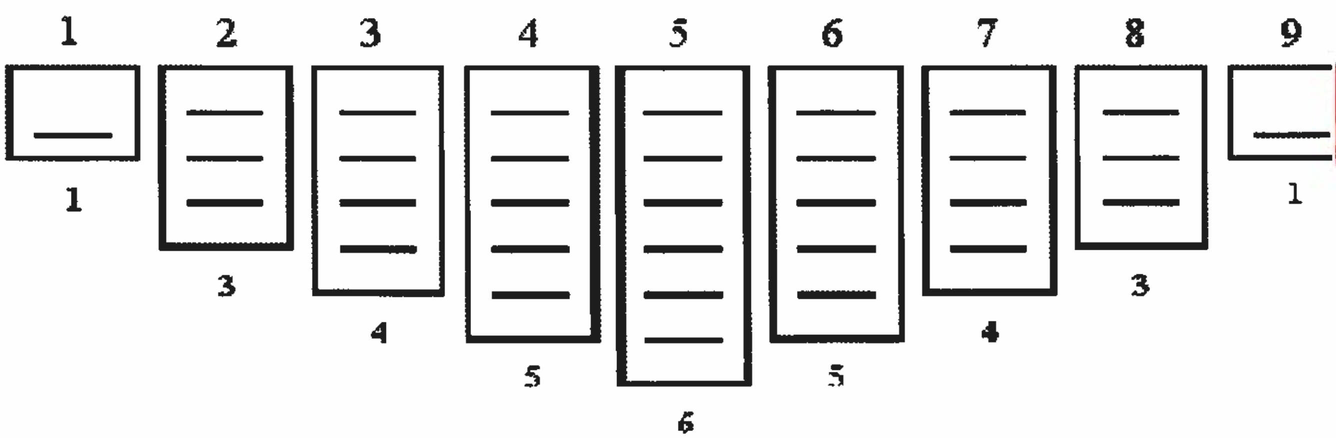
Charleston Southern University

ON-COURSE

Evaluation Analysis System

EVALUATION INPUT FORM

LEAST like Course **NEUTRAL or UNIMPORTANT** **MOST like**
Course



Instructions:

Arrange the "deck" of 32 cards into 9 piles according to the arrangement 1-3-4-5-6-5-4-3-1 as shown below. In other words, in the right-most, extremely characteristic pile (#9) place one and only one card, in the quite characteristic pile (#8) place three and only three cards, etc. When finished record the number of each card in the appropriate answer box provided below.

Pile No.	Label of Category
9	extremely characteristic of course
8	quite characteristic of course
7	fairly characteristic of course
6	somewhat characteristic
5	relatively neutral or unimportan
4	somewhat uncharacteristic
3	fairly uncharacteristic
2	quite uncharacteristic
1	extremely uncharacteristic

Charleston Southern University

ON-COURSE

Evaluation Analysis System

REQUEST REPORT

By Course | By Department | By Professor

Charleston Southern University

ON-COURSE

Evaluation Analysis System

COURSE UPDATE FORM

ADD COURSE		DELETE COURSE:		CHANGE COURSE INFO	
Name		Course #	Choose One	Course #	Choose One
Course #				Name	
Dept #				Dept #	
Instructor #				Instructor#	
Section #				Section #	
Add Course		Delete Course		Update Course	

will vary from semester to semester

Next Back

Only Courses for the current semester can be updated? Once a semester is completed I assume it moves to a "history" file and can't be changed!

Charleston Southern University

ON-COURSE

Evaluation Analysis System

DEPARTMENT UPDATE FORM

ADD DEPARTMENT:	DELETE DEPARTMENT:	CHANGE DEPARTMENT :
Dept #	Dept# Choose One	Dept # Choose One
Dept Name		Dept Name
Dept Chair		Dept Chair
Courses Included:		Courses Included:
Add Department	Delete Department	Update Department

Next Back

Not semester dependent?

Charleston Southern University

ON-COURSE

Evaluation Analysis System

PROFESSOR UPDATE FORM

ADD PROFESSOR	DELETE PROFESSOR	CHANGE PROFESSOR INFC
Name [ ]	Instructor # [Choose One]	Instructor # [Choose One]
Instructor # [ ]		Name [ ]
Dept # [ ]		Dept # [ ]
Phone # [ ]		Phone# [ ]
Courses Taught: [ ]		Courses Taught: [ ]
[Add Professor]	[Delete Professor]	[Update Professor]

[Next] [Back]

Cummulative? or for current semester only

Not semester dependent?

ON-COURSE

Evaluation Analysis System

EVALUATION INPUT FORM

Instructor # 146

Semester: Fall

Year: 2000

Course # 213

Department # 097

Section # 02

Next Back

EXAMPLE
INPUT
FILLED-OUT

ON-COURSE

Evaluation Analysis System

COURSE UPDATE FORM

ADD COURSE		DELETE COURSE:		CHANGE COURSE INFO	
Name	Business Financ	Course #	Choose One	Course #	Choose One
Course #	389			Name	
Dept #	097			Dept #	
Instructor #	294			Instructor#	
Section #	01			Section #	
Add Course		Delete Course		Update Course	

Next Back

OK
Press on

MILESTONE EVALUATION SHEET

MILESTONE NO.

11

DATE: 11/27/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

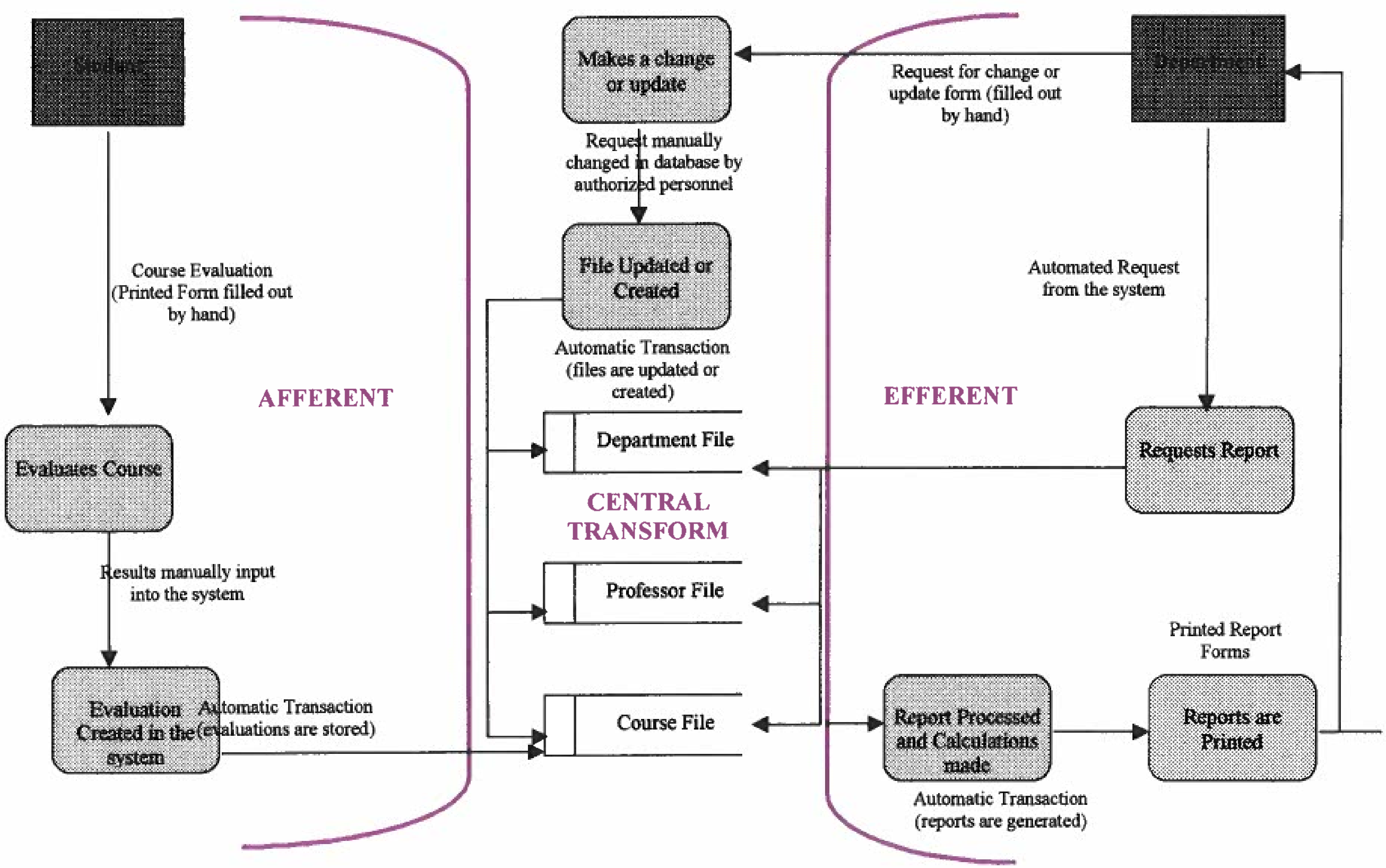
TEAM NO. OR NAME	TEAM MEMBER
Greenstone Greenstone CONSULTANTS	Karen Thurston

Student comments:

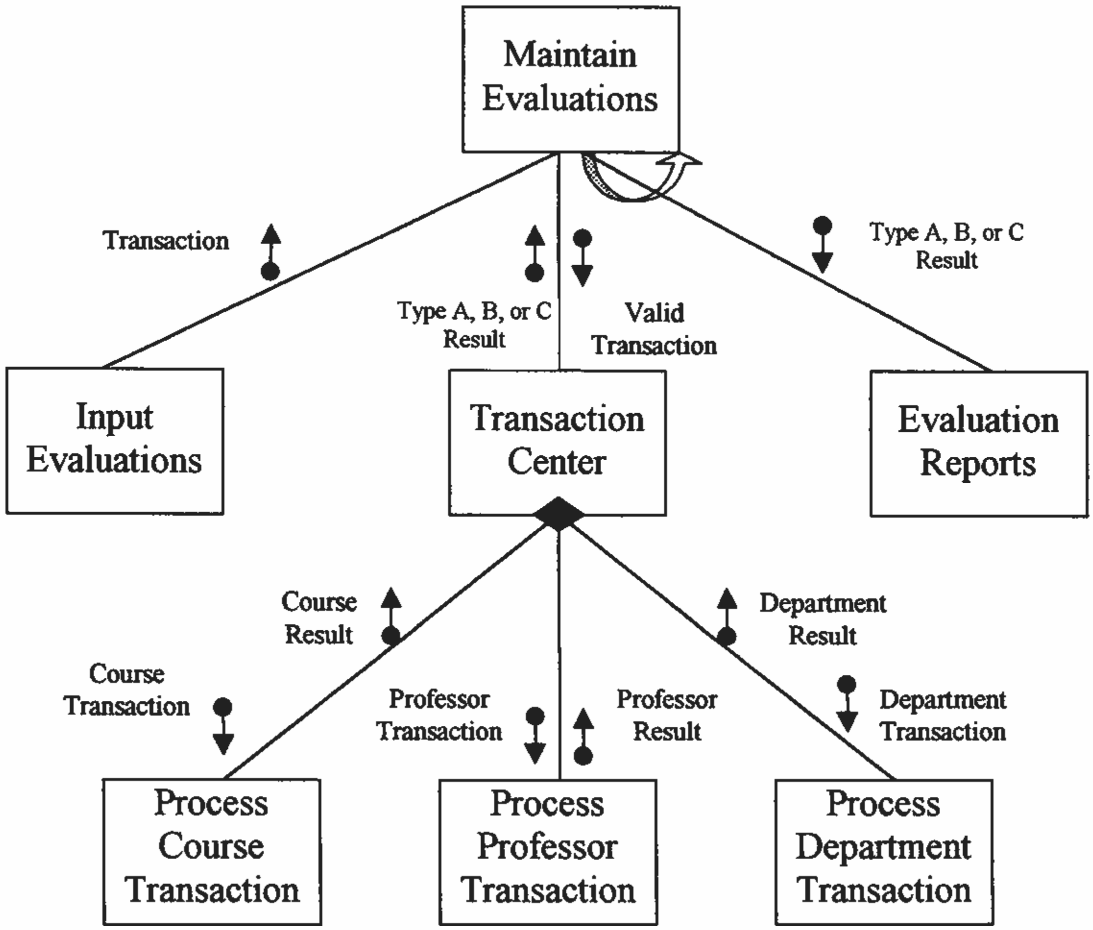
Instructor comments:

Score:

PARTITIONED DATA FLOW DIAGRAM



STRUCTURE CHART



MILESTONE EVALUATION SHEET

MILESTONE NO.

12

DATE: 12/7/2000	COURSE NO.: COINS 497 Senior Project Design
	INSTRUCTOR: Mr. Roberts

TEAM NO. OR NAME	TEAM MEMBER
Greenstone CONSULTANT	Karen Thurston

Student comments:

Instructor comments:

Tim, I want to be sure the program will have "flexibility." I need to ~~met~~ meet with you to explain what I mean by "flexibility."

Bill
12-8-00

Score:

GREENSTONE
CONSULTING

Dr. Bill Bowers
9200 University Blvd.
PO Box 118087
Charleston, SC 29423

December 7, 2000

Dr. Bowers,

Greenstone Consultants is very pleased to inform you that we have completed the analysis and design of a custom software package that will function as an evaluation analysis system. Greenstone provides the best, up-to-date technology designed to be user friendly as well as productive.

Your new system will have the capability to automate the reports that will provide accurate and complete information regarding course evaluations. The consistency of the automated program will provide detailed reports for all departments and professors upon request. Through proper maintenance your system will be dependable for years to come.

All objectives have been achieved within the specified time and budget constraints for this phase of analysis and design. Greenstone stands behind its products and services. Please review the attached report addressing the project in detail. If you have any questions regarding the system or any other concerns, please call feel free to contact Greenstone Consultants.

Greenstone has enjoyed working with you and look forward to continued business relationship in the future phases of this project.

Sincerely,

Karen Thurston

Karen Thurston
President
Greenstone Consultants

TECHNICAL REPORT

I. INTRODUCTION

A. Purpose

This report is designed to inform the programmers and those working in the next phases of this project of the progress and decisions made during the analysis and design of this particular project. It will summarize the background of the project as well as the work done during the analysis and design phase.

B. Background

Dr. Bill Bowers of Charleston Southern University has requested of Greenstone Consulting that an automated computer system be created that can take the students' evaluations and produce useful reports for the university and the individual departments. The current system is not set up to compare each course and professor to a standard set by the university, and therefore does not give as complete information about where the weaknesses are and which areas need most improvement. The statistics involved with the new system would give ~~the most~~ more complete and accurate information ~~possible~~ that could be used for improvements of the courses as well as the professors. The final system should be complete and implemented by May of 2001. The goal of this system is to make the courses and operations at Charleston Southern University more effective by providing an accurate evaluation of each course and professor by the students of the university.

Greenstone Consulting met with Dr. Bill Bowers of Charleston Southern University in September 2000. Dr. Bowers is the main contact for the project, and Karen Thurston is the project manager for Greenstone Consulting. Karen has been working with the assistance of Mr. Jim Roberts to design and implement this system as requested by Dr. Bill Bowers. The request was made by Dr. Bowers and evaluated by Karen Thurston. After evaluating the problems, opportunities, and objectives of this request, it was decided by Greenstone Consulting to go forth with this request. An allotted time of eight months has been given to the completion of this project, with the final deadline being May 2001. The analysis and design phase of this project has now been completed, and the project will soon proceed with the implementation and defense.

C. Scope

This project involves a system that will automatically evaluate the results of student course evaluations. The scope of the project at this time is to analyze the results of a new system of evaluations using Q-Methodology (McKeown and Thomas, 1988), and print reports requested by each department.

II. STANDARDS AND SPECIFICATIONS

A. Documentation Standards

The documentation used in the analysis and design phase was created using programs from Microsoft Office 2000.

B. Technical Environment (see Milestone 7)

The software created from this system should be able to be installed on any Pentium workstation with Windows 95 or higher. The workstations can store the software as well as the databases, or the databases can be housed on a central server. The workstations would require a connection to a LAN

IBM Laser #4019 Model

printer, preferably the ~~HP5SI LAN~~ Laser Printer. The software will be created using Microsoft Access, Microsoft Excel, and a Visual Basic front end.

III. SYSTEM OVERVIEW (Data Flow Diagrams)

A. Context, Decomposition, and System Diagrams (see Milestone 5)

B. Data Flow Diagram (see Milestone 8)

C. Partitioned Data Flow Diagram (see Milestone 11)

IV. FILE SPECIFICATIONS (all file and data base specifications)

Complete Table Definitions (see Milestone 9)

IV. PROGRAM SPECIFICATIONS

A. System Structure and Dialogue

Input Screens (see Milestone 10)

B. Program Specifications (Documentation of inputs, outputs, modular structure, and processing logic)

1. State Transition Diagram (see Milestone 10)

2. Sample Reports (see Milestone 10)

V. Implementation Schedule

A. Personnel Requirements

The head of each department will have the most interaction with this system and will be trained to use the program and request the desired reports. Personnel will be needed to input the results of the evaluations, which will be provided by each department. Greenstone will offer consulting and training upon request for a decided amount of time upon implementation of the system.

B. Schedule (Pert and/or Gantt Charts) (see Milestone 3)

VI. Recommendations and Conclusions

The system analysis and design phase of the requested project has been completed. The project may now progress into the design phase that will include building the system and the software and testing for operability. The next phases will commence in February 2001 and must be completed by May 2001.